

University of California, San Diego
Consent to Act as a Research Subject

Developing Novel Interactive Technology for Contextual and Activity Monitoring of
Human Behavior

Who is conducting the study, why you have been asked to participate, how you were selected, and what is the approximate number of participants in the study?

Dr. Edward Jay Wang of the University of California San Diego and his associates are conducting a research study to find out more about to develop new forms of interactive technology (digital pens, multitouch devices, depth-cameras, interactive walls, eye-tracking, wearable sensors, and computer interfaces) for monitoring context and activities of people. You have been asked to participate in this study because you fit the demographic that is representative of a user of the interactive technology we are studying. We are interested in understanding how these systems perform when used by you in a variety of natural and/or controlled laboratory situations. There will be a total of 500 participants in this study.

Why is this study being done?

Wearable sensors and mobile computing systems can help us understand your activity in terms of heart rate, perspiration, electrical activity, activities, and other important physiological measures. The purpose of this study is to improve our understanding of how digital devices and systems perform in realistic scenarios with actual human users, including how people interact with and interpret digital information, and to collect to improve these systems and validate performance.

What will happen to you in this study and which procedures are standard of care and which are experimental?

If you agree to be in this study, the following will happen to you:

This session will be conducted in a ☐ lab space ☐ public space ☐ remote or online.

In this session, the technologies that you will be using will be:

☐ Computer Interfaces	☐ Mobile Devices	☐ Multi-touch Devices
☐ Microphone (Stationary)	☐ Microphone (Wearable)	☐ Multimodal Digital Pens
☐ Display (Wall)	☐ Display (Wearable)	☐ Online survey or questionnaire
Camera (Stationary):	☐ Color ☐ IR ☐ Depth ☐ Thermal	
Camera (Wearable):	☐ Color ☐ IR ☐ Depth ☐ Thermal	
Noninvasive Wearable Sensor: ☐ Accelerometers ☐ Heart Rate Trackers ☐ EEG		
☐ EDA ☐ Respiration ☐ Eye Tracker ☐		

You will be asked to use the technology being examined in the experiment in a few everyday scenarios either scripted or asked to perform daily activities in as natural of way as possible. Activities can include such things as interacting with computer interfaces or mobile devices, changing body postures (standing up, sitting down, laying down, gesturing), non-strenuous exercise (walking, biking, pushups), daily patterns (using computer, eating or drinking, food preparation), and responding to information presented digitally, including reviewing textual information and answering questions. These daily activities are meant to mimic those what you would naturally do in your everyday life. For any activities that you have never performed or do not want to perform, the activity will either be removed for the session or replaced by an alternative activity that the participant has prior experience performing. The exact activity sequence you will perform and the duration of each has been provided to you. The staff will go over what each activity means with you.

The session will be conducted either in the presence of a study staff or remotely without direct staff interaction, depending on the study activity. settings. This will include the handling of equipment, instrumenting on the subject, and any announcement/handling of video/audio recording to bystanders to follow the legal requirements. For purposes of study staff review of the session, video and/or audio recording may be requested. If recording will occur, the study staff will inform you before starting the recording and obtain your consent for video and/or audio recording. If you do not feel comfortable with being recorded at any time, the staff will stop the recording. As long as the stopping of recording does not affect the study, you can continue to participate in the study.

How much time will each study procedure take, what is your total time commitment, and how long will the study last?

The study procedure for this session will be approximately ____ minutes as stated in the recruitment and will take no longer than 1 hour as maximum. The consent procedure will take about 5 minutes of the total time.

What risks are associated with this study?

Participation in this study may involve some added risks or discomforts. These include the following:

1. A potential for the loss of confidentiality. It is possible that some of the data collected will contain private and sensitive information; we will make every effort to maximize confidentiality such as keeping collected data securely stored, destroy field data if needed and ask explicit permission to publish personal images. Research records will be kept confidential to the extent allowed by law. Research records may be reviewed by the UCSD Institutional Review Board.
2. You might experience fatigue, discomfort or embarrassment using technology, but risks associated with using any of the technology introduced as part of this study are minimal and not higher than using those technologies in the day-to-day activities.
3. You might find out that using this technology can be overwhelming, intimidating, and/or frustrating. That's what we are studying, so remember that if something goes

wrong it is not your fault, but it is probably a failure of the technology designers to know what do the user want to do.

Because this is a research study, there may also be some unknown risks that are currently unforeseeable. You will be informed of any significant new findings.

What are the alternatives to participating in this study?

The alternatives to participation in this study is to not participate.

What benefits can be reasonably expected?

There may or may not be any direct benefit to you from participating this study. However, the results of this study may help in the design of new computer applications and interfaces in the future.

Can you choose to not participate or withdraw from the study without penalty or loss of benefits?

Participation in research is entirely voluntary. You may refuse to participate or withdraw or refuse to answer specific questions in an interview or on a questionnaire at any time without penalty or loss of benefits to which you are entitled. If you decide that you no longer wish to continue in this study, please inform the research staff and the study will be terminated.

You will be told if any important new information is found during the course of this study that may affect your wanting to continue.

Can you be withdrawn from the study without your consent?

The PI may remove you from the study without your consent if the PI feels it is in your best interest or the best interest of the study. You may also be withdrawn from the study if you do not follow the instructions given you by the study personnel.

Will you be compensated for participating in this study?

☐ No compensation	☐ \$____/hour	☐ Entry into a lottery or drawing for one of several gift cards valued at up to \$_____ each*
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*Note: entry into a lottery or drawing is not dependent on participation in the study. Individuals will be entered into the drawing even if they choose not to participate or withdraw from the study.

Are there any costs associated with participating in this study?

There will be no cost to you for participating in this study.

What if you are injured as a direct result of being in this study?

If you are injured as a direct result of participation in this research, the University of California will provide any medical care you need to treat those injuries. The University will not provide any other form of compensation to you if you are injured. You may call the Human Research Protections Program Office at 858-246-HRPP (858-246-4777) for more information about this, to inquire about your rights as a research subject or to report research-related problems.

Who can you call if you have questions?

Dr. Edward Jay Wang and/or _____ has explained this study to you and answered your questions. If you have other questions or research-related problems, you may reach Dr. Edward Jay Wang at (864) 719-9264.

You may call the Human Research Protections Program Office at 858-246-HRPP (858-246-4777) to inquire about your rights as a research subject or to report research-related problems.

What else should I know?

Dr. Edward Wang and Colin Barry have a potential conflict of interest as the cofounders of Billion Lab Inc.

Your Signature and Consent

You have received a copy of this consent document.

You agree to participate.

Subject's signature

Date